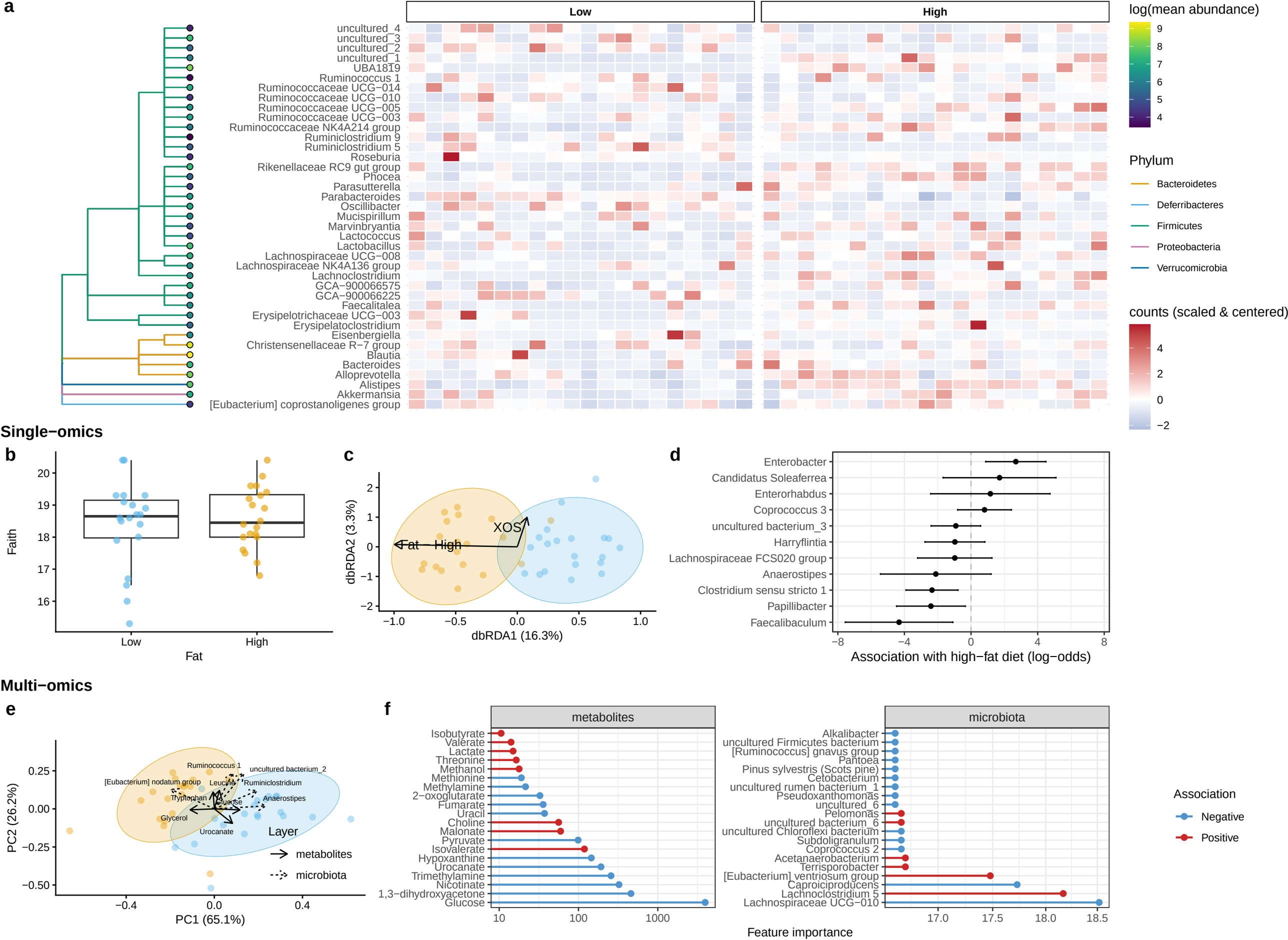
a
Low
High
uncultured_4
uncultured_3
uncultured_2
uncultured_1
UBA1819
Ruminococcus 1
Ruminococcaceae UCG−014
Ruminococcaceae UCG−010
Ruminococcaceae UCG−005
Ruminococcaceae UCG−003
Ruminococcaceae NK4A214 group
Ruminiclostridium 9
Ruminiclostridium 5
Roseburia
Rikenellaceae RC9 gut group
Phocea
Parasutterella
Parabacteroides
Oscillibacter
Mucispirillum
Marvinbryantia
Lactococcus
Lactobacillus
Lachnospiraceae UCG−008
Lachnospiraceae NK4A136 group
Lachnoclostridium
GCA−900066575
GCA−900066225
Faecalitalea
Erysipelotrichaceae UCG−003
Erysipelatoclostridium
Eisenbergiella
Christensenellaceae R−7 group
Blautia
Bacteroides
Alloprevotella
Alistipes
Akkermansia
[Eubacterium] coprostanoligenes group
log(mean abundance)
9
8
7
6
5
4
Phylum
Bacteroidetes
Deferribacteres
Firmicutes
Proteobacteria
Verrucomicrobia
counts (scaled & centered)
4
2
0
−2
Single−omics
b
Faith
20
19
18
17
16
Low
High
Fat
c
dbRDA2 (3.3%)
2
1
0
−1
−2
XOS
Fat − High
−1.0
−0.5
0.0
0.5
1.0
dbRDA1 (16.3%)
d
Enterobacter
Candidatus Soleaferrea
Enterorhabdus
Coprococcus 3
uncultured bacterium_3
Harryflintia
Lachnospiraceae FCS020 group
Anaerostipes
Clostridium sensu stricto 1
Papillibacter
Faecalibaculum
−8
−4
0
4
8
Association with high−fat diet (log−odds)
Multi−omics
e
PC2 (26.2%)
0.25
0.00
−0.25
−0.50
Ruminococcus 1
uncultured bacterium_2
[Eubacterium] nodatum group
Leucine
Ruminiclostridium
Tryptophan
Anaerostipes
Glucose
Glycerol
Urocanate
Layer
metabolites
microbiota
−0.4
0.0
0.4
PC1 (65.1%)
f
metabolites
Isobutyrate
Valerate
Lactate
Threonine
Methanol
Methionine
Methylamine
2−oxoglutarate
Fumarate
Uracil
Choline
Malonate
Pyruvate
Isovalerate
Hypoxanthine
Urocanate
Trimethylamine
Nicotinate
1,3−dihydroxyacetone
Glucose
10
100
1000
microbiota
Alkalibacter
uncultured Firmicutes bacterium
[Ruminococcus] gnavus group
Pantoea
Pinus sylvestris (Scots pine)
Cetobacterium
uncultured rumen bacterium_1
Pseudoxanthomonas
uncultured_6
Pelomonas
uncultured bacterium_6
uncultured Chloroflexi bacterium
Subdoligranulum
Coprococcus 2
Acetanaerobacterium
Terrisporobacter
[Eubacterium] ventriosum group
Caproiciproducens
Lachnoclostridium 5
Lachnospiraceae UCG−010
17.0
17.5
18.0
18.5
Feature importance
Association
Negative
Positive